Metodologias de Desenvolvimento de Software 2025/2026

Mestrado em Informática e Sistemas

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Software Development Methodologies

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1. Outline

What's this course about?

"Software Development Methodologies"

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Objectives

- The main objective of this course is to provide knowledge, understanding and practice on different software development methodologies
  - Software life-cycle and software processes
  - Traditional and agile development methodologies
  - Software maturity

Learning Outcomes

- On completion of this course, the students shall be able to:
  - A1. Identify the different phases of a software life cycle
  - A2. Know different agile software development methodologies
  - A3. Understand software maturity levels and goals
  - B1. Develop small applications according to agile development principles
  - B2. Develop small applications according to different traditional development principles
  - C1. Evaluate the need for formal requirements
  - C2. Justify the applicability of different methodologies to specific projects
  - C3. Analyze real development projects
  - E1. Write simple and correct artifacts

Course Philosophy

- Theoretical classes with presentation of the subjects and study materials. Discussion of case studies.
- Out of the class work
  - Study
  - Assignments for feedback
  - Graded assignments
- Labs with
  - Discussion of the assignments
  - Discussion of other case studies
  - Practical work

Contents

1. Motivation
   - Case studies on software development failures
   - Examples of software development approaches
2. Software Process
   - Process, framework, lifecycle
3. Process Models
   - ISO/IEC 12207, CMMI, ISO/IEC 15504
4. Traditional methodologies
   - Waterfall, incremental, evolutionary
5. Agile methodologies
   - Scrum, Extreme Programming, Kanban, Agile Unified Process, Scrum at Scale
6. Design Thinking
7. Data Science Methodologies
8. AI tools
   - BOTSchool, Lovable
9. Conclusion
   - Global analysis

Assessment

- 5 assignments:
  - Executive summaries based of the analysis of the case studies, experiences, studies
  - Presentation and arguing
  - 1 assignment will be totally graded during class
  - total 13 points
- Assignment Submission
  - A student may present 2 assignments at the last class (17/12)
  - A student may submit 2 assignments until 22/12
    - This is only valid for assignment 1,3 and 5.
    - Presentation and submission are mandatory for the assignments

Assessment

- Final exam
  - Covering all the subjects
  - 7 points
  - A minimum of 30% is mandatory for passing

5 assignments

Optional:

Case studies- Software Failures (17/Sep) – 0 pts

Mandatory:

1. Agile *vs* traditional methodologies (07/Oct or 08/Oct) – 2.5 pts
2. Scrum in Practice (21/Oct or 22/Oct) – 0.5 pt (full in-class evaluation)
3. Scaling Scrum (28/Oct or 29/Oct) – 2.5 pts
4. Practice with agile methodologies – teamwork (05/Nov to 03/Dec) – 5pts
5. Data Science Project Methodologies (09/Dec or 10/Dec) – 2.5 pts

Last class (17/Dec) – Presentation (and submission) of overdue work

22/Dec –Submission of overdue work

Working Hours

- Total: 191 hours
- Per week
  - +/- 4 hours class
  - +/- 4 hours out-of-class work
- Per assignment
  - +/- 12 hours work out of the classes (more for assignment 4)
- Study for final exam
  - 45 hours

Class Timeframe

- Some weeks may have 3 hours (more practical) instead of 4 hours (theory + lab)

- At these weeks the schedule will be:
  - P1 (normal Schedule)
  - T
    - From 18:00 to 19:00 to P1
    - From 19:00 to 20:00 to P2
  - P2 (normal Schedule)

  - 15/10/2025 – 3 hours
  - 22/10/2025 – 3 hours
  - 12/11/2025 – 3 hours (assignment 4)
  - 19/11/2024 – 3 hours (assignment 4)
  - 26/11/2024 – 3 hours (assignment 4)
  - 03/12/2024 – 3 hours (assignment 4)

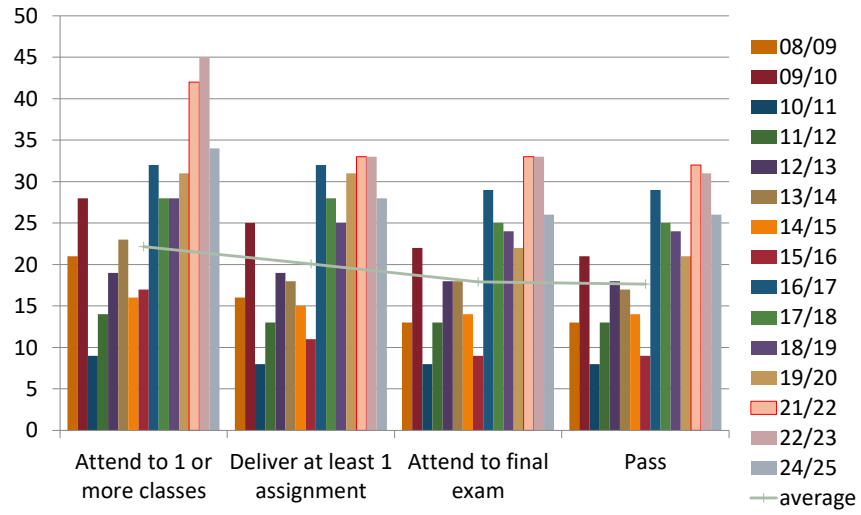
Bibliography (always in progress...)

- Tsui, Frank, and Orlando Karam. *Essentials of Software Engineering.* Sudbury, MA: Jones and Bartlett Publishers, 2007.
- Pressman, Roger S. *Software Engineering: a practitioner's approach - 6th ed.* New York: McGraw-Hill, 2005.
- Abrahamsson, Pekka, Outi Salo, Jussi Ronkainen, and Juhani Warsta. *Agile Software Development Methods - Review and Analysis.* Finland: VTT Publications, 2002.
- Shore, James, and Shane Warden. *The Art of Agile Development.* O'Reilly, 2007.
- Several papers.

Support

- All the material available at Microsoft Teams (https://teams.microsoft.com/)
  - Also **inforestudante.ipc.pt** when necessary (e.g. summaries)
- Communication between instructor and students preferably using General chat or direct contacts at MS Teams
- Assignments delivered at MS Teams
- Grades and feedback to students through MS Teams

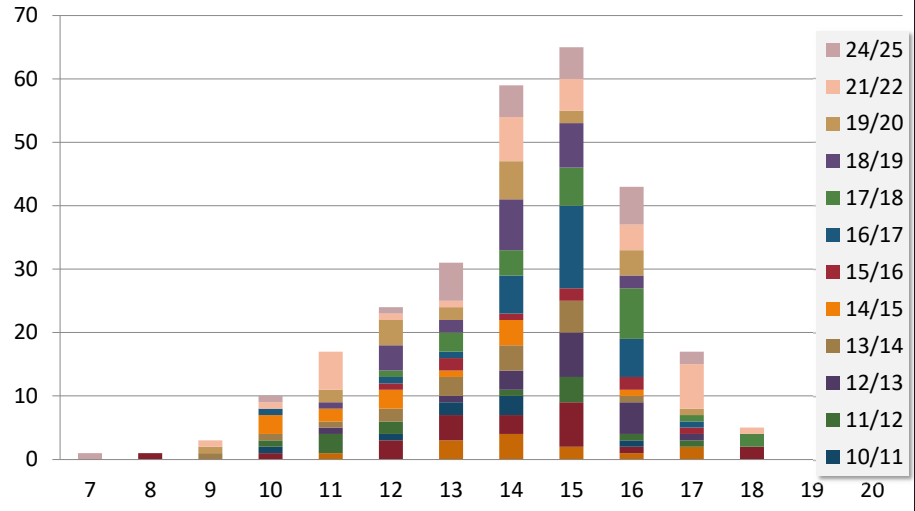
Students' success 2008/2009 - 2024/2025



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Grades 2008/2009 until 2024/2025



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